

THOTH Project

Subsystem Division Plan

Research & Learning Phase Structure

Overview

This document presents a structured and balanced approach to dividing the THOTH project into three main subsystems. The purpose of this division is to:

- Allow all team members to explore different technical areas
- Ensure fair workload distribution during the research and documentation phase
- Build a shared understanding across all subsystems
- Maintain flexibility for future implementation adjustments

At this stage, the focus is on learning, researching, and prototyping, not strict specialization. Each team will later share their findings with the rest of the group to ensure that everyone gains sufficient knowledge across the system.

Subsystem 1: Navigation & Robot Control

Main Goal

Enable the robot to move safely and efficiently within the museum environment.

What this subsystem includes

- Robot movement pipeline
- Environment mapping
- Localization within the map
- Path planning between exhibits
- Real-time obstacle avoidance
- Sensor integration and control logic

Main Tasks

- Robot model setup and motion configuration
- SLAM (Simultaneous Localization and Mapping)
- Localization techniques
- Path planning algorithms
- Obstacle detection and avoidance
- Motor control interface
- Simulation of movement before real deployment

Requirements

- Understanding of ROS 2 fundamentals
- Working with simulation environments and robot nodes

- Testing in simulated environments
- Sensor integration (LiDAR, depth cameras)
- Coordination with GUI for map visualization

Tools & Technologies

- ROS 2
- Nav2
- Gazebo
- RViz
- SLAM Toolbox
- LiDAR systems
- AprilTags (optional)
- Micro-ROS / ESP32 communication

Importance

This subsystem is one of the core pillars of the project, as it enables the robot to physically guide visitors and navigate autonomously within the museum.

Subsystem 2: AI Interaction & Exhibit Intelligence

Main Goal

Enable intelligent interaction between the robot and visitors through natural communication.

What this subsystem includes

- Visitor question handling
- Museum knowledge base creation
- Natural language conversation
- Multilingual support (Arabic & English)
- Speech input/output integration
- Optional visual understanding (future extension)

Main Tasks

- Structuring exhibit data
- Designing a Q&A system
- Implementing multilingual responses
- Integrating speech-to-text and text-to-speech
- Connecting AI models with curated knowledge
- Reducing hallucinations using grounded data
- Testing and refining responses

Requirements

- Organizing and structuring information effectively
- Building backend logic or APIs
- Testing accuracy and response quality
- Coordination with GUI for displaying responses
- Coordination with navigation for contextual responses

Tools & Technologies

- LLM APIs (e.g., OpenAI or alternatives)
- Whisper (or other STT tools)
- Text-to-Speech systems
- Python
- Flask / FastAPI
- JSON / Databases
- RAG (optional later)
- Optional CV tools: OpenCV / YOLO / MediaPipe

Importance

This subsystem enables the robot to provide interactive, intelligent, and personalized experiences, making it a key feature of THOTH.

Subsystem 3: GUI, Multimedia & Visitor Experience

Main Goal

Create an intuitive and engaging interface for user interaction.

What this subsystem includes

- Touchscreen interface
- Exhibit information display
- Multimedia content (images, videos, possibly 3D)
- Accessibility features (captions, controls)
- Navigation interface and map display
- User interaction flow

Main Tasks

- Designing UI layouts and user experience
- Building exhibit display screens
- Creating navigation selection interfaces
- Displaying AI responses visually

- Implementing captions and accessibility options
- Supporting bilingual interface
- Integrating with AI and navigation systems

Requirements

- UI/UX design understanding
- Frontend development
- Content management and presentation
- Backend/API integration
- Simplicity and usability for real users

Tools & Technologies

- Figma
- React / Web frontend or PyQt
- HTML / CSS / JavaScript
- Flask / FastAPI integration
- JSON / Database storage
- Multimedia handling tools
- Optional map visualization

Importance

This subsystem ensures that all system capabilities are accessible, usable, and engaging for visitors, making it essential for the user experience.

Why This Division Works

This structure is balanced because each subsystem provides:

- A clear technical focus
- A distinct set of deliverables
- A real implementation path
- Both research and practical components

Workload Balance:

- Navigation → Algorithm-heavy and simulation-focused
- AI → Logic, backend, and intelligent processing
- GUI → Interface, design, and system integration

Each subsystem is challenging in a different way, ensuring fairness.

Team Distribution (Research Phase)

To maintain fairness and equal learning:

- 2 members → Navigation & Robot Control
- 2 members → AI Interaction & Exhibit Intelligence
- 2 members → GUI, Multimedia & UX

Inside each subsystem:

- 1 member focuses on research & architecture
- 1 member focuses on prototype & implementation

Collaboration & Flexibility

This division is not rigid.

- The current phase focuses on learning and exploration
- Each team will share knowledge and findings with others
- Regular discussions ensure that all members understand all subsystems

Later in implementation:

- The structure can be adjusted based on workload
- If navigation becomes more complex → more members can join
- If another subsystem requires support → redistribution is possible

This ensures:

- No one is locked into a single role
- Everyone has exposure to all system components
- The team remains flexible and adaptive

Current Deliverables (Research Phase)

Each subsystem should produce:

1. Research Summary

- Subsystem overview
- Available approaches
- Chosen methods and justification

2. Tools Learning Summary

- Technologies explored
- Setup progress
- Learning resources

3. Small Prototype

- Navigation → basic simulation
- AI → simple Q&A system
- GUI → interface mockup or working screen

4. Risks & Challenges

- Potential issues
- Proposed solutions or alternatives

Subsystem Selection Guide

Choose based on your interest:

- Navigation & Robot Control → Robotics, simulation, algorithms, motion systems
- AI Interaction & Exhibit Intelligence → Backend, AI, APIs, logic, data systems
- GUI, Multimedia & UX → Design, frontend, user experience, visualization

Conclusion

This structure ensures:

- Fair workload distribution
- Strong technical learning
- Flexibility for future implementation
- Shared understanding across the team

Structured for Learning, Built for Flexibility 